

Department of Computer Science
Gurucharan University, Silchar
3rd Semester C++ Practical Questions
(Part1)

Q1. Define a class Rectangle with data members length and breadth. Implement member functions to:

- Read the dimensions.
- Calculate and display the area and perimeter of the rectangle.

Create an object of the class and demonstrate its use.

Q2. Define a class Vector to represent a two-dimensional vector with components x and y. Implement member functions to:

- Input the vector components.
- Add two vectors.
- Display the resultant vector.

Demonstrate vector addition using objects.

Q3. Create a class Bank with data members such as account number, account holder name, and balance. Implement member functions for:

- Deposit
- Withdrawal
- Displaying account details

Ensure that withdrawal is allowed only when sufficient balance is available.

Q4. Write a C++ program to create a class Student containing roll number, name, and marks. Use a constructor to initialise the data members and a member function to display the student details.

Q5. Write a C++ program to demonstrate call by value and call by reference using suitable functions. Demonstrate the effect of modifying values using both techniques.

Q6. Define a class Calculator and implement function overloading to perform addition of:

- Two integers
- Three integers
- Two floating-point numbers

Create an object and demonstrate all overloaded functions.

Q7. Write a C++ program to demonstrate the implementation of a static data member and a static member function. Create a class Employee with a static data member to count the number of objects created. Display the total number of objects using a static member function.

Q8. Define two classes ClassA and ClassB, each containing a private data member. Implement a friend function that accesses the private data members of both classes and displays their sum.

Q9. Create a class Distance with data members feet and inches. Write a member function that accepts an object of the same class as a parameter and adds two distance objects. Display the resultant distance in proper feet and inches format.

Q10. Create a class Employee with appropriate data members. Implement the following:

- A constructor to initialise employee details.
- A member function to calculate salary.
- A static data member to count the number of employees.
- A static member function to display the employee count.
- A friend function to compare the salary of two employee objects.

Create multiple objects and demonstrate all the above concepts.